Installing and Calibrating a 3D Touch / BLTouch on the Longer LK5 Pro

Field-tested guide for the MKS Robin E3 V1.1 controller and DWIN touchscreen

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> **Read the compatibility and firmware-format warnings before flashing anything.** The Longer LK5 Pro was sold with more than one controller family. This guide documents one successfully tested STM32-based MKS Robin E3 V1.1 installation. It is **not** a guide for the older Robin Nano V3.5 or AVR-controller printers.

1. What this guide covers

This manual follows a real installation from initial wiring and a failed firmware update through successful recovery, 3D Touch testing, Z-offset calibration, 6 × 6 bed mapping, Cura configuration, and a five-square first-layer test. It includes the diagnostic commands, expected output, and failure checks that proved useful.

Successfully verified on the tested machine: deploy/retract, probe-trigger sensing, two-pass Z homing, a 36-point bilinear mesh, EEPROM storage, Cura mesh activation, and five first-layer squares adhering across the 300 × 300 mm bed.

Not guaranteed: other controller revisions, alternative 3D Touch clones, different bootloaders, display variants, or firmware revisions may need other settings. Avoid blind flashing.

2. Hardware identification — do this first

The tested machine used:

Item	Tested configuration
Printer	Longer LK5 Pro; 300 × 300 × 400 mm
Controller	**MKS Robin E3 V1.1**
MCU	STM32F103RCT6, 256 KiB flash, 48 KiB RAM
Screen	Longer DWIN 4.3-inch serial touchscreen
Probe	3D Touch / BLTouch-compatible five-wire probe
Probe servo pin	**PB1** via board connector **J12**
Probe trigger input	**PC6 / Z-MIN** via connector **J11**
Mechanical Z switch	Disconnected on tested installation
Host system	Ubuntu 24.04, PlatformIO, CuteCom
Serial communication	115200 baud, 8N1, flow control off

Open the electronics enclosure **only after disconnecting power**. Read the actual silkscreen and PCB revision. A model number on the printer's external label is insufficient.

Do not use a firmware image for a Robin Nano V3.5 or other controller simply because its archive says "LK5 Pro." An older package named `LK5PRO BL.rar` was identified during the project as the incompatible `0.4.0-BLDual` firmware generation.

3. Tools and prerequisites

- Five-wire 3D Touch / BLTouch and a rigid mount suited to the LK5 Pro toolhead.
- Multimeter for checking power and ground, wire order, and continuity; consult the board silkscreen and probe manufacturer's wiring documentation.
- FAT32 microSD card; USB cable; Ubuntu/Linux laptop or other build-capable computer.
- Python virtual environment and PlatformIO; Git and a serial terminal (CuteCom was used).
- A clean sheet of ordinary printer paper, plus a way to stop printer power promptly during first Z testing.
- A backup of **all** original configuration settings (send `M503` from the working machine and save the complete response). If available, preserve an independently verified, known-working firmware image.

Important: a file renamed `FIRMWARE.OLD` by the SD updater may be the uploaded firmware renamed by the bootloader, **not a backup of the firmware originally running**. Do not assume it is safe recovery firmware.

4. Electrical connections

For the **tested E3 V1.1 board**:

Probe function	Board connection	MCU signal
5 V, ground, servo/control	J12, appropriate 3-pin positions	PB1 control
Trigger output and trigger return	J11, Z-MIN input pair	PC6 trigger

Wire colors and connector pin positions vary among probe clones. Verify the probe's actual pin assignments and the board's labels before powering on. Reversing 5 V and ground can damage hardware. Do **not** infer pin order from the connector housing or another user's wire colors.

The mechanical Z-min switch was disconnected in the tested configuration, because the probe now supplies the Z-min trigger. Never home Z until the probe responds to both the control and trigger tests below.

Fit the probe rigidly. With its pin **extended**, the probe tip must project below the nozzle tip, so it can contact the bed first. Check clearance throughout X/Y travel, including around bed clips.

Repository wiring picture:
https://github.com/DreadedZombie/LK5Pro-1.0.3E3-Marlin2.X-Firmware/blob/main/LK5%20Pro%20BLtou

5. Firmware sources, compatibility, and the hard-earned lesson

The reference repository was:

https://github.com/DreadedZombie/LK5Pro-1.0.3E3-Marlin2.X-Firmware

It contains a Marlin 2.0.x LK5 Pro configuration and a precompiled reference archive:

`FDM-LK5PRO-FW-1.0.3E3-BL-230905.zip`

On the tested printer, the archive's `firmware.bin` was linked at application address `0x08005000` and contained **unencrypted ARM Thumb code**. Its SHA-256 was:

`c992f410136cc724747434a8cb13d472d0a41974f8210a2885f7a011aba23791`

The cloned project's build generated **two distinct files**:

- `.pio/build/mks_robin_e3_maple/firmware.bin` — **raw, unencrypted** STM32 application.
- `.pio/build/mks_robin_e3_maple/Robin_e3.bin` — **XOR-transformed** by the MKS/Robin post-build script.

The repository README instructed renaming `Robin_e3.bin` to `firmware.bin` before flashing. **Following that advice produced a bootloader update that appeared successful but left the touchscreen on the LONGER splash screen and the application unresponsive.** A second encrypted build linked at `0x08007000` also failed on this particular machine.

The diagnosis changed when we compared the reference firmware with both outputs. At offset `0x140` (320 bytes), the unencrypted files contained `00 80 01 23 23 70 10 BD`, whereas the encrypted image began `A3 3D AC 2E 62 61 AB 30`. The reference and successful custom build also had the same reset-handler address `0x080053A1`.

Confirmed result for the tested bootloader: the **raw `firmware.bin` linked for `0x08005000` booted**, with the BLTouch enabled. Do not universalize this result to other MKS bootloaders. A firmware flash progress bar reports writing, not necessarily successful execution or compatibility.

The working customized image built during the project had 183,404 bytes. Its SHA-256 was:

`8bcd9056a9f443a084d66595dd692705597e65f49e43d6aa48ee658add2ae30e`

Recovery rule: verify hardware revision, source, format, reset vector, and update instructions first. Preserve known-working images. When uncertain, **do not flash**.

6. Building Marlin on Ubuntu 24.04

Clone the exact project (or an audited fork) and create an isolated PlatformIO environment:

```
git clone https://github.com/DreadedZombie/LK5Pro-1.0.3E3-Marlin2.X-Firmware.git LK5Pro-BLTo
cd LK5Pro-BLTouch
python3 -m venv .venv
.venv/bin/python -m pip install --upgrade pip
.venv/bin/python -m pip install platformio
```

Check the project's configuration; these were the relevant tested selections:

- `LGT`, `LGT_LCD_DW`, `LK5_PRO` printer/display definitions.
- `BOARD_MKS_ROBIN_E3_V1_1` motherboard.
- `WITH_Z_PROBE` enabled, with Marlin's `BLTOUCH`, `Z_MIN_PROBE_USES_Z_MIN_ENDSTOP_PIN`, `AUTO_BED_LEVELING_BILINEAR`, and `Z_SAFE_HOMING` enabled through the project's conditional configuration.
- `MULTIPLE_PROBING 2`; 6 × 6 mesh.
- `NOZZLE_TO_PROBE_OFFSET { -45, -20, 0 }` **as the starting X/Y estimates** for this specific mount. Another mount must be measured separately. The Z offset is calibrated later, not guessed.
- Probe margin was changed from 80 to 30 mm in this project. Confirm the physical reach and bed clips before `G29`.

Boot address check (these are the three files that were examined after a failed attempt):

```
buildroot/share/PlatformIO/scripts/mks_robin_e3.py
buildroot/share/PlatformIO/ldscripts/mks_robin_e3.ld
ini/stm32f1-maple.ini    [env:mks_robin_e3_maple]
```

The **working configuration** used `0x08005000` in the Robin prepare call, linker ROM origin and upload-offset setting. The appropriate script is `mks_robin_e3.py`, not the similarly named Robin E3P or Robin Nano scripts.

Build:

```
.venv/bin/pio run -e mks_robin_e3_maple -t clean
.venv/bin/pio run -e mks_robin_e3_maple
```

Check for `SUCCESS`. The reference setup used PlatformIO STM32 platform 12.1.1, Maple Arduino framework 3.10000.201129, and ARM GCC 7.2.1. Other package combinations should be tested rather than assumed equivalent.

Verify output before placing it on an SD card

```
xxd -l 32 .pio/build/mks_robin_e3_maple/firmware.bin
readelf -h .pio/build/mks_robin_e3_maple/firmware.elf | grep 'Entry point'
sha256sum .pio/build/mks_robin_e3_maple/*.bin
```

On the tested successful build the first eight bytes of the raw binary encoded initial SP `0x2000BFD8` and reset handler `0x080053A1`. Never treat matching vectors alone as proof that a firmware image is safe for a given board.

7. Flashing the firmware

Before updating: save `M503` settings and a copy of every firmware binary on your computer. Check the SD card's filesystem and free space. Remove unrelated firmware update files.

For the **tested controller and bootloader only**, copy the **raw** build artifact to the FAT32 card's root using the **lowercase exact name** `firmware.bin`:

```
cp .pio/build/mks_robin_e3_maple/firmware.bin /path/to/MOUNTED-SD/firmware.bin
sync
```

Replace the mount path with your real card path; unmount/eject safely before inserting it into the powered-off printer. Do not blindly substitute `Robin_e3.bin`.

Power on. A successful updater may report:

```
initialize flash...ok
erase flash...ok
flash firmware...ok
update done.
start application.
```

`start application` is not proof of a boot. Wait for the Marlin screen and/or a serial `M115` response. One successful recovery took a short pause before the normal DWIN menu appeared. When the update completes, power off, remove the SD card and restart; verify the display and serial response again.

On Linux, identify the USB adapter port with:

```
ls -l /dev/serial/by-id/
ls -l /dev/ttyUSB* /dev/ttyACM* 2>/dev/null
id -nG
```

The tested USB serial adapter appeared as `/dev/ttyUSB0` and used **115200 baud, 8N1, no flow control**. CuteCom sometimes requires explicitly selecting

the serial device even when its correct name is initially displayed. The SD card is not needed for ordinary serial communication.

Test:

```
M115
M503
```

If EEPROM reports an index/data-size mismatch after a firmware change, preserve the old settings first. `M502` loads **new firmware defaults** and `M500` saves them, which overwrites stored calibration values. Re-enter machine-specific adjustments from the saved `M503` record as appropriate.

8. Verify the probe without homing

Place the nozzle safely above the bed. Confirm the LED is not rapidly flashing (alarm). On the tested configuration:

```
M280 P0 S160 ; alarm reset, if needed
M280 P0 S10  ; deploy pin
M280 P0 S90  ; stow pin
```

A normal deploy/stow sequence occurred, and a rapidly flashing red alarm cleared to steady red following reset.

To check the trigger input, deploy the pin and query endstops:

```
M280 P0 S10
M119
```

Expected with pin **extended and untouched**: `z_min: open`.

Gently lift the pin to trigger it, without forcing or holding it if the probe retracts automatically, and request `M119` again. Expected **after trigger**: `z_min: TRIGGERED`.

Do **not** home Z if these states are reversed, stuck, unreliable, or if the probe is in alarm. Filament-runout status is separate; the tested machine reported `filament: TRIGGERED` and the sensor should be investigated independently if printing is unexpectedly blocked.

9. Safe homing and probe positioning

With clearance confirmed and power switch accessible:

```
G28 X Y
G28 Z
M114
```

Check the printhead moves to a safe bed location, the probe extends and **stops downward Z movement on contact**, and the toolhead rises for a second

pass where configured. Be ready to stop power if the nozzle approaches the bed while the probe does not respond.

The tested offsets were **X -45, Y -20**. The printer reported nozzle X195, Y170 during safe Z homing, placing the **probe** around X150, Y150 (center of the bed).

For safety, never regard a displayed Z coordinate as physical clearance until the probe Z offset has been calibrated. Do not issue `G1 Z0` based solely on a successful homing message.

10. Z-offset paper calibration

This section describes the method actually used, not a universal starting offset. On the tested machine the starting probe offset was `M851 X-45 Y-20 Z0`; the final first-layer-tuned Z offset was **-2.22 mm**.

1. Home the printer successfully with the probe. Move the **nozzle**, not just the probe, over the bed center (the tested target was nozzle X150 Y150).
2. Place ordinary printer paper under the nozzle. Approach conservatively: large moves only when clearance is ample, smaller moves near the bed. The tested sequence went from reported Z10 to Z5, then Z0, followed by submillimeter increments. The observed physical gap at Z0 was still several millimeters because the offset was initially zero.
3. `M211` checks software endstops. When `M211 S1` and minimum Z0 prevented movement below the provisional origin, software endstops were **temporarily disabled** using `M211 S0`, with careful 0.5 mm, then 0.1 mm moves and constant observation. **This is an advanced, potentially bed-damaging step.** Use the printer's supported Z-probe-offset wizard if available instead. Never move into the bed or continue if you feel strong paper resistance.
4. At reported Z **-2.20**, the paper began to drag lightly, producing the starting value `M851 Z-2.20`. Restore `M211 S1` **immediately**. Save with `M500`.
5. Rehome and verify with software endstops **ON**. On the tested printer, slight paper contact near Z0 confirmed the starting estimate. Two subsequent five-square prints were carried out with Z offset **-2.22 mm**, which gave well-adhered, flexible first-layer sheets.
6. Never assume a paper test precisely measures final print height: paper thickness, nozzle cleanliness, bed temperature, filament flow, and texture matter. Use first-layer printing for small, deliberate fine adjustment.

Commands used to save and confirm the final field-tested Z offset:

```
M851 X-45 Y-20 Z-2.22
M211 S1
M500
M503
M211
```

Do not copy -2.22 mm for a different printer or mount; determine the probe-to-nozzle vertical offset for that specific installation.

11. Generate and save the 6 × 6 mesh

First mechanically tram the bed, warm it to typical printing temperature for best relevance, and home all axes. The tested print bed had been manually leveled before starting.

```
G28
G29
```

Wait for `G29` to complete. The printer may send many `echo:busy: processing` lines while probing. If it replies `echo:Home XYZ First`, perform a **fresh** successful `G28` and immediately retry `G29` without resetting the printer.

The tested mesh spanned from **-0.162 mm** to **+0.208 mm**, a **0.370 mm** peak-to-peak variation. This variation was handled well in the five-square printing test, but another machine will have different values.

After `G29` completes without errors:

```
M500      ; Save settings / mesh
M420 S1   ; Enable bed leveling
M420 V    ; Verify status and print mesh
```

The tested printer reported `Bed Leveling ON` and fade height `4.00`.

Depending on firmware, `G28` may disable leveling even though the mesh remains stored. **Always reactivate the saved mesh after homing in Cura Start G-code.** Verify mesh state with `M420 V` after significant changes.

12. Cura Start G-code

In Cura: **Settings → Printer → Manage Printers → select Longer LK5 Pro → Machine Settings → Start G-code**. Preserve your existing purge lines and other printer-specific startup behavior. Place `M420 S1 Z4` **immediately after** `G28`:

```
; LONGER Start G-code - tested layout
G21 ; metric values
G90 ; absolute positioning
M82 ; set extruder to absolute mode
M107 ; start with the fan off
G92 E0 ; Reset Extruder
G28 ; Home all axes
M420 S1 Z4 ; Enable saved 6x6 BLTouch mesh, fade over 4 mm
G1 Z2.0 F3000 ; Lift nozzle
G1 X0.1 Y20 Z0.3 F5000.0 ; Move to purge start
G1 X0.1 Y200.0 Z0.3 F1500.0 E15 ; First purge line
```

```
G1 X0.4 Y200.0 Z0.3 F5000.0 ; Move sideways
G1 X0.4 Y20 Z0.3 F1500.0 E30 ; Second purge line
G92 E0 ; Reset Extruder
G1 Z2.0 F3000 ; Lift nozzle
G1 X5 Y20 Z0.3 F5000.0 ; Move away
```

Confirm Cura produces temperature/wait commands appropriate to your material before cold extrusion is attempted. Slice a model, open the resulting `.gcode` file, and ensure the generated startup really contains `G28` followed by `M420 S1 Z4`.

13. First-layer validation

Print **five single-layer, approximately 30 × 30 mm squares**: center plus four corners of the 300 × 300 mm bed, with reasonable margin from bed clips. Use initial-layer height 0.20 mm, 100% infill, two walls, approximately 15–25 mm/s first-layer speed, and no supports or adhesion brim for this diagnostic. Verify the sliced preview is **one layer**. The exact model placement and shape should be validated in Cura before printing.

Good results: squares remain attached, extrusion lines are joined, and cooled squares peel away as continuous flexible sheets. An excessively high nozzle leaves separated strands; an overly low nozzle leaves scraping, severe ridges, or flattened/translucent thin spots. Slight translucency **alone** does not establish a wrong Z offset; textured glass imprints and optics also contribute.

In the tested installation, all five squares adhered, peeled as flexible sheets, and showed only minor local appearance variations; the last two photographic sets used **Z -2.22 mm**. The bed mesh remained unchanged. We did **not** establish precise film thickness because the available caliper resolution was insufficient to reliably distinguish 0.1 from 0.2 mm locally.

14. Troubleshooting quick reference

Symptom	Checks and actions
Update reports 100% but printer stays at LONGER logo	Verify controller revision, boot offset, encoding, and recovery file; `start application` alone is not proof. This machine required **raw** `firmware.bin` at `0x08005000` rather than the XOR-transformed `Robin_e3.bin`.
No serial response	Confirm actual `/dev/serial/by-id/` device, baud 115200, 8N1, no flow control, explicit device selection in CuteCom. Test `M115`, `M105`.

Symptom	Checks and actions
EEPROM data-size/index error	Save old settings; `M502` then `M500` resets/stores new defaults, but loses previous stored calibration unless restored.
Probe flashes red rapidly	Check physical pin, wiring, obstruction; `M280 P0 S160` alarm reset; verify deploy and stow before any motion.
Z-min reads triggered at rest	Deploy the probe, then check `M119`; expect `open` while untouched and `TRIGGERED` when activated.
No movement from `G29`	Look for `Home XYZ First`; run `G28` then `G29` while powered and homed.
`G29` approaches clip/edge	Stop; review probe offset, X/Y usable limits, margin, mount reach and bed placement.
`G1 Z-1` returns `ok` but doesn't move	`M211` may show software Z minimum 0; do not repeatedly command into the limit. Use a controlled, safe offset procedure.
Mesh saved but not used in print	Insert `M420 S1 Z4` **after** `G28` in Cura start code; verify `M420 V`.
Local first-layer translucency	Check actual extrusion-line bonding, bed texture, nozzle gap, filament flow and thermal conditions before changing the global Z offset.

15. Useful links and further reading

LK5 Pro and firmware

- Controller-specific Marlin repository: https://github.com/DreadedZombie/LK5Pro-1.0.3E3-Marlin2.X-Firmware
- Reference firmware ZIP in that repository: https://github.com/DreadedZombie/LK5Pro-1.0.3E3-Marlin2.X-Firmware/blob/main/FDM-LK5PRO-FW-1.0.3E3-BL-230905.zip
- Probe connector photograph: https://github.com/DreadedZombie/LK5Pro-1.0.3E3-Marlin2.X-Firmware/blob/main/LK5%20Pro%20BLtouch%20Pin%20Locations.png
- DWIN display notes (project-specific, not a universal flashing procedure): https://github.com/DreadedZombie/LK5Pro-1.0.3E3-Marlin2.X-Firmware/blob/main/DWIN_UI_CUSTOMIZATION.md
- PlatformIO installation: https://docs.platformio.org/en/latest/core/installation/index.html

Marlin G-code documentation

- `M119` endstop states: https://marlinfw.org/docs/gcode/M119.html
- `M280` servo/probe pin commands: https://marlinfw.org/docs/gcode/M280.html
- `M851` probe offsets: https://marlinfw.org/docs/gcode/M851.html
- `G29` bilinear mesh: https://marlinfw.org/docs/gcode/G029-abl-bilinear.html
- `M420` leveling state and fade height: https://marlinfw.org/docs/gcode/M420.html
- `M500` save settings: https://marlinfw.org/docs/gcode/M500.html
- `M502` restore defaults: https://marlinfw.org/docs/gcode/M502.html

Slicing

- UltiMaker Cura: https://ultimaker.com/software/ultimaker-cura/

16. Reproducibility notes and scope

This manual reflects a **single successful user installation** and specific terminal outputs, not a controlled test of all hardware variants. The most important finding is that the conventional "use the encrypted Robin image" instruction was **not applicable to the tested bootloader**. The working file was the **raw STM32 `firmware.bin`**, built at `0x08005000`; the custom firmware with 3D Touch was confirmed to boot by its 2026 compile timestamp and Marlin serial startup output.

Never publish an image as a universal recovery firmware solely because it booted one printer. Keep firmware artifacts, Git revision, configuration headers, build logs, SHA-256, and a working EEPROM settings record together for future reproducibility.

Prepared from an actual October 2026 LK5 Pro + 3D Touch troubleshooting, recovery, and calibration session. Names of products and linked software remain the property of their respective owners. This guide is independently authored; the linked firmware retains its upstream GPL licensing.